

Answer Key — Biology Grade 10

Sub-Strand 1.2: Chemicals of Life • Lesson 5: Sensemaking: Connecting the Chemistry of Food to Health Outcomes in Kisumu

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
A	D	B	C	C	B	B

USE AFTER: PREDICT PHASE — Predict: individual prediction sentence on the most important chemical difference between ugali and githeri

Q1. Two children in a village in Kisumu eat the same amount of food each day. One eats only ugali (maize starch). The other eats githeri (maize and beans). Based on food test results, what is the most important chemical difference between ugali and githeri?

- A. Githeri contains protein as well as carbohydrate, but ugali is mainly carbohydrate**
- B. Ugali contains protein, but githeri does not
- C. Ugali contains more food than githeri, so it gives less energy
- D. Githeri contains no carbohydrates, but ugali does

Correct answer: A. Ugali is mainly starch (a carbohydrate), while githeri combines maize (carbohydrate) with beans (protein). The difference is the type of chemical, not the quantity of food.

USE AFTER: OBSERVE PHASE — Observe: Deficiency Disease Matching Card, citing the Biuret test result for kwashiorkor

Q2. In a Biuret test, a bean extract turns purple and a sample of ugali stays blue. What does this result show?

- A. Beans contain starch and ugali does not
- B. Ugali contains more lipids than beans
- C. Beans contain vitamin C and ugali does not
- D. Beans contain protein, while ugali has little or no protein**

Correct answer: D. The Biuret test gives a purple colour when protein is present. Beans give a purple result and ugali does not, so ugali is a poor source of protein.

USE AFTER: OBSERVE PHASE — Observe: diet audit of the daily menus for the two Kisumu children

Q3. In a diet audit, Child 1 eats ugali three times a day. Child 2 eats githeri twice and ugali once. Which statement about their diets is correct?

- A. Child 2's diet supplies protein but no carbohydrates
- B. Child 1's diet supplies carbohydrates but too little protein, and Child 2's diet supplies both**
- C. Child 1's diet has no carbohydrates, but Child 2's diet has carbohydrates
- D. Both diets supply the same chemicals of life because they contain the same amount of food

Correct answer: B. Ugali supplies starch (carbohydrate) but little protein. Githeri adds protein from beans, so Child 2 gets both carbohydrate and protein.

USE AFTER: OBSERVE PHASE — Observe: Deficiency Disease Matching Card, completing the row for scurvy

Q4. A learner tests orange juice with DCPIP in the lab and it changes colour. A disease is caused by a lack of this same chemical in the diet. Which disease is it?

- A. Marasmus
- B. Kwashiorkor
- C. Scurvy**
- D. Rickets

Correct answer: C. DCPIP tests for vitamin C, which orange juice contains. A lack of vitamin C causes scurvy.

USE AFTER: EXPLAIN PHASE — Explain: written group explanation using the sentence starter on albumin, keratin and muscle fibres

Q5. A child with kwashiorkor has a swollen belly, thin limbs and discoloured hair. Which row correctly links a symptom to the protein that is lacking?

- A. Swollen belly: keratin loss
- B. Thin limbs: keratin loss
- C. Swollen belly: albumin loss**
- D. Discoloured hair: albumin loss

Correct answer: C. Albumin helps keep fluid in the blood, so too little albumin causes oedema (swollen belly). Keratin makes hair, and muscle protein makes muscle, so its loss causes thin limbs.

USE AFTER: EXPLAIN PHASE — Explain: whole-class discussion evaluating each group's explanation of the anchoring phenomenon

Q6. Which statement is the best scientific explanation for why the child who eats only ugali develops kwashiorkor, while the child who eats githeri stays healthy?

- A. The ugali child does not get enough energy because ugali has no carbohydrates
- B. The ugali child gets enough energy from starch but too little protein for building body tissues**
- C. The githeri child gets more vitamins, and vitamins alone build muscle and hair
- D. The ugali child eats a smaller quantity of food than the githeri child

Correct answer: B. Ugali gives energy from carbohydrate, but the body needs protein (amino acids) to build tissues such as albumin, keratin and muscle. The problem is a lack of protein, not a lack of food quantity.

USE AFTER: MODEL BUILDING PHASE — Model: partial A3 'Our Chemicals of Life Model' template, filling in the protein box

Q7. A group is filling in the protein box of their 'Our Chemicals of Life Model' template. Which entry is correct?

- A. Food source: orange juice; consequence of deficiency: rickets
- B. Food source: beans; consequence of deficiency: kwashiorkor**
- C. Food source: ugali; consequence of deficiency: scurvy
- D. Food source: ugali; consequence of deficiency: anaemia

Correct answer: B. Beans are a Kenyan source of protein, and a lack of protein causes kwashiorkor. Ugali is mainly carbohydrate, and scurvy, rickets and anaemia are linked to other missing chemicals.